

Statistics & Probability Theory

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1 A/B Testing, A/B/n Testing, and Multivariate Testing

A/B testing is a randomized controlled experiment used to compare two variants of a system to determine which performs better with respect to a predefined metric such as conversion rate or revenue per user. A/B/n testing extends this framework to multiple variants, while multivariate testing decomposes system performance into contributions from multiple factors and their interactions.

1.1 Experimental Design and Group Assignment

The first step is to define the objective and primary metric (e.g., conversion rate). Users are filtered into the target population (e.g., new users), and then randomly assigned to groups

To ensure balance across important covariates (e.g., geography, device type), stratified randomization is used. The population is divided into strata and randomization is applied within each stratum.

Let strata be indexed by s , then:

$$P(\text{assignment} \mid s) = \text{uniform across variants}$$

This reduces variance and ensures comparability across groups.

1.2 Sanity Checks

Before analysis, sanity checks are required:

- Sample Ratio Mismatch (SRM): verify observed allocation matches expected proportions

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- Balance checks: compare distributions of covariates across groups
- Data quality checks: ensure event logging consistency

Failure in these checks invalidates the experiment.

1.3 Statistical Significance Testing

Let p_A and p_B be conversion rates of control and variant. The null hypothesis is:

$$H_0 : p_A = p_B$$

A common test is the two-sample z-test:

$$z = \frac{p_A - p_B}{\sqrt{p(1-p) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}$$

where p is the pooled estimate.

1.4 Numeric Example (A/B Testing)

- Variant A: $n_A = 10000$, conversions = 500 $\Rightarrow p_A = 0.05$
- Variant B: $n_B = 10000$, conversions = 650 $\Rightarrow p_B = 0.065$

Pooled estimate:

$$p = \frac{500 + 650}{20000} = 0.0575$$

$$z = \frac{0.065 - 0.05}{\sqrt{0.0575(1 - 0.0575) \left(\frac{1}{10000} + \frac{1}{10000} \right)}} \approx 4.2$$

This corresponds to a very small p-value, indicating statistical significance.

Statistical significance indicates the probability of observing data under the null hypothesis. A p-value below a threshold (e.g., 0.05) suggests rejecting H_0 . However, significance does not imply practical importance.

1.5 A/B/n Testing

A/B/n testing compares multiple variants simultaneously. Traffic is split across k variants.

This lab demonstrates an A/B/n testing framework with five variants (A, B, C, D, E), where A serves as the control group and the remaining groups represent different treatment variants. The goal is to evaluate whether changes in design or strategy lead to improvements in user behavior.

The dataset consists of 300 observations, with 60 users assigned to each group. Each observation contains a binary conversion outcome (0 or 1) and a revenue value, where revenue is non-zero only for converted users. This structure reflects a realistic experimental setup commonly used in product and marketing analytics.

The primary metric considered is the conversion rate, defined as the proportion of users who completed the desired action. A secondary metric is revenue, which captures the monetary value generated per user.

Group	Users	Conversions	Conversion Rate	Avg Revenue
A	60	17	0.283	29.0
B	60	35	0.583	58.3
C	60	40	0.667	65.0
D	60	45	0.750	75.5
E	60	48	0.800	84.2

Table 1: Summary statistics for each group

From the summary statistics, the conversion rates across the groups are as follows: A: 0.283, B: 0.583, C: 0.667, D: 0.750, E: 0.800. There is a clear increasing trend from the control group to the final variant, suggesting that each successive treatment improves performance.

To formally test whether these differences are statistically significant, a multi-sample test for equality of proportions was conducted. The null hypothesis states that all groups have equal conversion rates, while the alternative hypothesis states that at least one group differs.

The test produced a p-value of 1.544×10^{-8} , which is far below the significance level of $\alpha = 0.05$. Therefore, the null hypothesis is rejected, indicating that there is a statistically significant difference in conversion rates across the groups.

	A	B	C	D
B	0.0174	-	-	-
C	0.0006	1.0000	-	-
D	0.0000081	0.8136	1.0000	-
E	0.00000039	0.1769	1.0000	1.0000

Table 2: Pairwise comparison p-values (Bonferroni adjusted)

To identify which specific groups differ, pairwise comparisons were performed using a Bonferroni correction to control for multiple testing. The results show that:

- Group A differs significantly from all other groups.
- Differences among groups B, C, D, and E are mostly not statistically significant after correction.
- The largest improvements occur when moving from the control (A) to any treatment group.

Comparison	Diff	Lower CI	Upper CI	Adjusted p-value
B - A	29.27	5.14	53.39	0.0086
C - A	36.05	11.93	60.17	0.0005
D - A	46.54	22.42	70.67	0.0000023
E - A	55.17	31.05	79.29	0.0000000
C - B	6.78	-17.34	30.91	0.9386
D - B	17.28	-6.85	41.40	0.2855
E - B	25.90	1.78	50.03	0.0284
D - C	10.49	-13.63	34.62	0.7549
E - C	19.12	-5.00	43.24	0.1920
E - D	8.63	-15.50	32.75	0.8635

Table 3: Tukey HSD results for revenue comparison

In addition to conversion, revenue was analyzed as a continuous outcome using one-way ANOVA. The null hypothesis states that the mean revenue is equal across all groups. The ANOVA test yielded a p-value of 9.83×10^{-9} , indicating strong evidence against the null hypothesis and confirming that revenue differs significantly across groups.

A Tukey post-hoc test was conducted to determine which pairs of groups show significant differences in mean revenue. The results indicate that:

- All treatment groups (B, C, D, E) generate significantly higher revenue than the control group (A).
- Differences among higher-performing groups are less pronounced, with only a few pairwise comparisons reaching significance.

Overall, both conversion rate and revenue metrics consistently improve from group A to group E. While the global tests confirm statistically significant differences, pairwise comparisons suggest diminishing marginal gains among the top-performing variants.

From a business perspective, group E demonstrates the highest conversion rate and average revenue, making it the strongest candidate for deployment. However, since differences between top variants are not always statistically significant, considerations such as implementation cost, user experience, and robustness should also be taken into account before making a final decision.

2 Poisson Distribution

The Poisson distribution is used to model the **number of events occurring in a fixed interval of time or space**.

- Events occur randomly and independently
- The average rate of occurrence is constant
- We count how many events occur in a fixed interval

We denote:

$$X \sim \text{Poisson}(\lambda), \quad \lambda > 0$$

- λ is the **rate parameter**
- Represents the expected number of events in a fixed interval
- Example: $\lambda = 5$ means on average 5 events per interval
- Larger $\lambda \Rightarrow$ more frequent events
- Smaller $\lambda \Rightarrow$ fewer events

2.1 Probability Mass Function (PMF)

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

2.2 Cumulative Distribution Function (CDF)

$$P(X \leq k) = \sum_{i=0}^k \frac{\lambda^i e^{-\lambda}}{i!}$$

2.3 Mean and Variance

$$\mathbb{E}[X] = \lambda$$

$$\text{Var}(X) = \lambda$$

Interpretation:

- Mean = Variance (key property)
- Larger λ increases both spread and center

2.4 Derivation of the Mean

$$\begin{aligned} \mathbb{E}[X] &= \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k e^{-\lambda}}{k!} \\ &= \lambda \end{aligned}$$

2.5 Derivation of Variance

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$\text{Var}(X) = \lambda$$

2.6 Connection to Exponential Distribution

- Poisson models **counts of events**
- Exponential models **waiting time between events**
- If events follow a Poisson process:
 - Inter-arrival times are Exponential
 - Number of events in interval is Poisson

2.7 Shape and Behavior

- For small λ : right-skewed
- For large λ : approximately normal
- Distribution shifts right as λ increases
- The MLE for λ is the sample mean
- Empirical and theoretical probabilities should align well
- Mean and variance should be approximately equal
- Deviations indicate overdispersion or model mismatch

2.8 Model Diagnostics

- Check if mean $\approx$ variance
- If variance $>$ mean $\Rightarrow$ overdispersion
- If variance $<$ mean $\Rightarrow$ underdispersion
- Consider Negative Binomial if Poisson fails

2.9 Applications

- Number of calls in a call center per hour
- Number of emails received per day
- Number of accidents at an intersection
- Count of rare events in fixed intervals

3 Exponential Distribution

The exponential distribution is used to model the **waiting time until the occurrence of an event** in a continuous-time setting.

- Events occur randomly over time
- The average rate of occurrence is constant
- We are interested in the time until the first event occurs

We denote:

$$X \sim \text{Exponential}(\lambda), \quad \lambda > 0$$

- λ is called the **rate parameter**
- Think of λ as Expected number of events per unit time
- For $\lambda = 0.2$, the process generates about 0.2 events per unit time, i.e., on average about 2 events in 10 time units.
- The rate is time-independent: events occur randomly over time, but with a stable average rate that does not change with time.
- Larger $\lambda \Rightarrow$ faster events $\Rightarrow$ shorter waiting times
- Smaller $\lambda \Rightarrow$ slower events $\Rightarrow$ longer waiting times

3.1 Probability Density Function (PDF)

The probability density function is given by:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

Properties:

- $f(x) \geq 0$
- $\int_0^\infty f(x) dx = 1$

Interpretation: The density $f(x)$ describes how likely the event is to occur around time x .

3.2 Cumulative Distribution Function (CDF)

$$F(x) = P(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt$$

$$F(x) = 1 - e^{-\lambda x}$$

Interpretation: The probability that the event has occurred by time x .

3.3 Survival Function

$$S(x) = P(X > x) = 1 - F(x) = e^{-\lambda x}$$

Interpretation: The probability that the event has not yet occurred by time x .

3.4 Mean and Variance

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

Interpretation:

- The expected waiting time is inversely proportional to the rate
- Higher rate $\Rightarrow$ smaller mean waiting time

3.5 Derivation of the Mean

$$\mathbb{E}[X] = \int_0^\infty x \lambda e^{-\lambda x} dx$$

Using integration by parts:

Let:

$$u = x, \quad dv = \lambda e^{-\lambda x} dx$$

$$du = dx, \quad v = -e^{-\lambda x}$$

Then:

$$\mathbb{E}[X] = [-x e^{-\lambda x}]_0^\infty + \int_0^\infty e^{-\lambda x} dx$$

$$= 0 + \frac{1}{\lambda}$$

$$\Rightarrow \mathbb{E}[X] = \frac{1}{\lambda}$$

3.6 Derivation of Variance

The variance is given by:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

First compute:

$$\mathbb{E}[X^2] = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx$$

Using integration by parts:

$$\begin{aligned}\mathbb{E}[X^2] &= 2 \int_0^{\infty} x e^{-\lambda x} dx \\ &= \frac{2}{\lambda^2}\end{aligned}$$

Thus:

$$\text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

3.7 Memoryless Property

The exponential distribution satisfies the memoryless property:

$$P(X > s + t \mid X > s) = P(X > t)$$

Interpretation:

- The future waiting time does not depend on how much time has already passed
- The process has no memory

Proof:

$$\begin{aligned}P(X > s + t \mid X > s) &= \frac{P(X > s + t)}{P(X > s)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} \\ &= P(X > t)\end{aligned}$$

3.8 Shape and Behavior

- The density function is highest at $x = 0$ and decreases exponentially
- The distribution is right-skewed
- Most events occur early, with a long tail for larger values
- Higher λ implies shorter expected waiting time:

$$\mathbb{E}[T] = \frac{1}{\lambda}$$

- For a fixed time t , a higher λ increases the probability that the event occurs early:

$$P(T \leq t) = 1 - e^{-\lambda t}$$

- Thus, larger λ leads to faster occurrence of events and higher probability of occurrence within a given time interval.

4 Survival Analysis

Survival analysis deals with modeling the time until an event of interest occurs. Common examples include time to death, failure of a machine, or customer churn.

A key feature of survival data is the presence of censoring, where the event of interest is not observed for all individuals within the study period.

4.1 Survival Function

The survival function is defined as:

$$S(t) = P(T > t)$$

Interpretation: The probability that the event has not occurred by time t .

4.2 Hazard Function

The hazard function is defined as:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

Interpretation: The instantaneous rate at which events occur at time t , given survival up to time t .

4.3 Kaplan–Meier Estimator

The Kaplan–Meier estimator is a non-parametric estimator of the survival function:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

where:

- n_i = number at risk just before time t_i
- d_i = number of events at time t_i
- KM → estimates survival directly.
- KM is not a “first step in modeling” — it’s a separate non-parametric estimator

Time	2	3	6	6	7	10	15	15	16	27	30	32
Event	1	0	1	1	1	0	1	1	1	1	1	1

Note: 1 indicates event occurred, 0 indicates right censoring.

Time	n_i	d_i	$\hat{S}(t)$	Std. Err	Lower 95% CI	Upper 95% CI
2	12	1	0.917	0.0798	0.7729	1.000
6	10	2	0.733	0.1324	0.5148	1.000
7	8	1	0.642	0.1441	0.4132	0.996
15	6	2	0.428	0.1565	0.2089	0.876
16	4	1	0.321	0.1495	0.1287	0.800
27	3	1	0.214	0.1325	0.0635	0.720
30	2	1	0.107	0.1005	0.0169	0.675
32	1	1	0.000	NaN	NA	NA

At time $t = 2$:

$$\hat{S}(2) = 1 \times \left(1 - \frac{1}{12}\right) = \frac{11}{12} = 0.917$$

At time $t = 6$:

$$\hat{S}(6) = 0.917 \times \left(1 - \frac{2}{10}\right) = 0.917 \times 0.8 = 0.733$$

At time $t = 7$:

$$\hat{S}(7) = 0.733 \times \left(1 - \frac{1}{8}\right) = 0.733 \times 0.875 = 0.642$$

The Kaplan–Meier estimator is also known as the product-limit estimator:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

This arises from decomposing the survival probability into conditional probabilities:

$$P(T > t) = \prod_{t_i \leq t} P(T > t_i | T \geq t_i)$$

Each term represents the probability of surviving past time t_i given survival up to that point

Thus, the overall survival probability is obtained by multiplying these conditional survival probabilities across all observed event times.

Censored observations:

- Contribute to the risk set (n_i)
- Do not contribute to event count (d_i)

Example:

- At time 3 and 10, observations are censored
- They reduce future risk sets but do not affect survival probability directly
- The survival curve is a step function
- Drops occur only at event times
- Larger drops indicate higher event rates
- Increasing uncertainty over time due to fewer individuals at risk

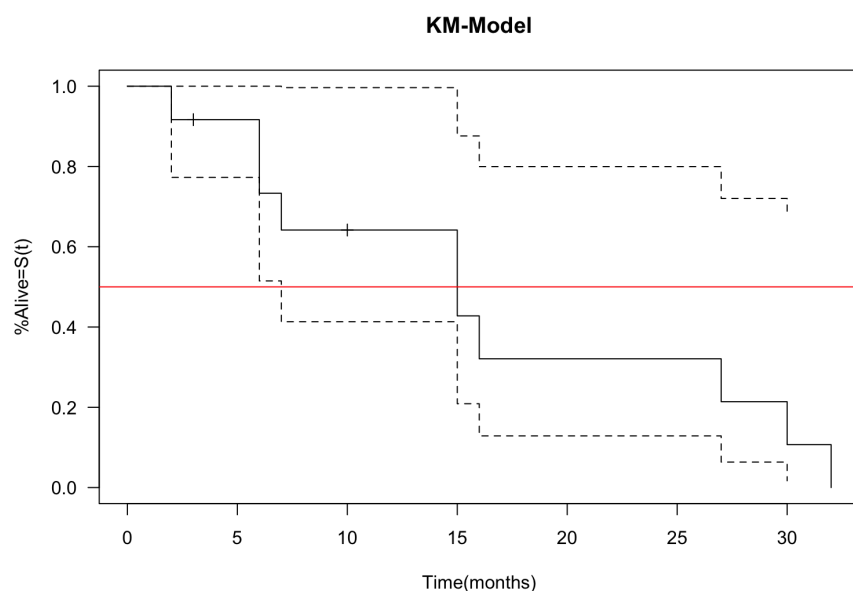


Figure 1: Kaplan–Meier survival curve

The median survival time is defined as the smallest time t such that:

$$\hat{S}(t) \leq 0.5$$

In this dataset, the survival probability drops below 0.5 at $t = 15$, hence the median survival time is 15.

Interpretation: Approximately 50% of the individuals experience the event by time 15.

4.4 Survival Analysis (Kaplan-Meier Estimator) with Two Treatment Groups

We consider a dataset of 20 patients undergoing treatment for a disease. Patients are divided into two groups:

- Treatment A (standard)
- Treatment B (new treatment)

For each patient, we observe:

- Time to event (in months)
- Event indicator (1 = death, 0 = censored)
- Treatment group

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Time	2	3	6	6	7	10	15	15	16	27	4	5	8	9	12	14	18	20	22	25
Status	1	0	1	1	1	0	1	1	1	1	1	1	0	1	0	1	0	1	1	0
Treatment	A	A	A	A	A	A	A	A	A	A	B	B	B	B	B	B	B	B	B	B
Age	65	70	72	68	66	74	69	71	67	73	60	62	59	63	61	64	58	66	65	67

Time	n_i	d_i	$\hat{S}(t)$	Std.Err	Lower CI	Upper CI
2	10	1	0.900	0.0949	0.7320	1.000
6	8	2	0.675	0.1551	0.4303	1.000
7	6	1	0.563	0.1651	0.3165	1.000
15	4	2	0.281	0.1631	0.0903	0.876
16	2	1	0.141	0.1286	0.0234	0.844
27	1	1	0.000	NaN	NA	NA

Time	n_i	d_i	$\hat{S}(t)$	Std.Err	Lower CI	Upper CI
4	10	1	0.900	0.0949	0.7320	1.000
5	9	1	0.800	0.1265	0.5868	1.000
9	7	1	0.686	0.1515	0.4447	1.000
14	5	1	0.549	0.1724	0.2963	1.000
20	3	1	0.366	0.1884	0.1332	1.000
22	2	1	0.183	0.1600	0.0329	1.000

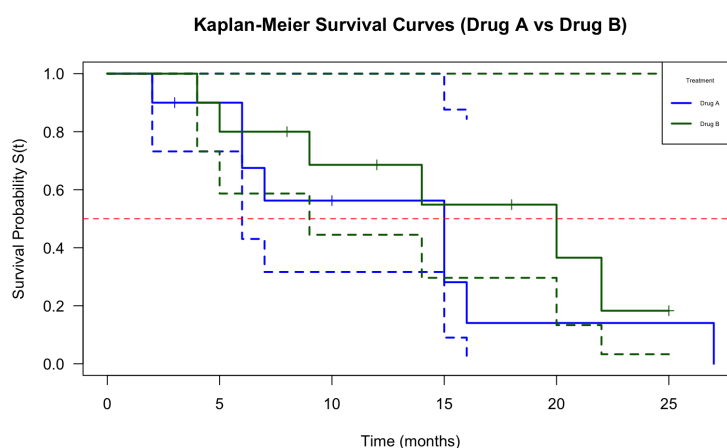


Figure 2: Kaplan–Meier Survival Curves for Treatment A and B

- Treatment B generally shows higher survival probabilities at later times.
- Treatment A experiences more rapid decline in survival.
- The difference between curves is visible but not very large.

Median Survival

- Treatment A reaches survival below 0.5 earlier.
- Treatment B maintains survival above 0.5 for longer.
- Interpretation: Patients receiving Treatment B tend to survive longer.

Statistical Significance (Log-Rank Test)

- Null hypothesis: Survival curves are identical.
- Test statistic: $\chi^2 = 0.6$
- p-value = 0.4
- Since $p = 0.4 > 0.05$, we fail to reject the null hypothesis.
- There is **no statistically significant difference** between the two treatment groups.

4.5 Exponential Model (Intro to Regression Models for Survival)

The exponential model assumes a constant hazard rate:

$$h(t) = \lambda, \quad \lambda > 0$$

$$f(t) = \lambda e^{-\lambda t} \quad (\text{PDF: density at time } t)$$

$$F(t) = 1 - e^{-\lambda t} \quad (\text{CDF: } P(T \leq t), \text{ probability event occurs by time } t)$$

$$S(t) = e^{-\lambda t} \quad (\text{Survival: } P(T > t), \text{ probability of surviving beyond } t)$$

The key assumption of the exponential model is: The hazard is constant over time.

- The risk of failure in the next instant does not depend on how long the individual has survived
- This is known as the **memoryless property**

To allow different individuals to have different risks, we model:

$$\begin{aligned} h(t | X) &= \lambda(X) \\ h(t | X) &= \exp(X\beta) \end{aligned}$$

- Ensures $h(t | X) \geq 0$
- Provides a linear structure on the log scale:

$$\log h(t | X) = X\beta$$

- Coefficients are interpretable:
 - e^{β_j} represents the multiplicative effect on the hazard

Each individual follows an exponential distribution:

$$T_i \sim \text{Exponential}(\lambda_i), \quad \lambda_i = \exp(X_i\beta)$$

- Same functional form across individuals
- Different hazard levels depending on covariates

$$S(t | X) = \exp(-e^{X\beta} \cdot t)$$

- Larger $X\beta$ leads to faster decay of survival
- Smaller $X\beta$ leads to slower decay

Likelihood Derivation for Exponential Regression Model Let T_i denote the survival time for individual i , and let $\delta_i \in \{0, 1\}$ be the event indicator, where $\delta_i = 1$ if the event is observed and $\delta_i = 0$ if the observation is right-censored. Let t_i be the observed time.

For an exponential model with rate λ_i , the density and survival functions are:

$$f(t_i | \lambda_i) = \lambda_i e^{-\lambda_i t_i}, \quad S(t_i | \lambda_i) = P(T_i > t_i) = e^{-\lambda_i t_i}.$$

The likelihood contribution depends on what is observed:

- If $\delta_i = 1$, the exact event time is observed, so the contribution is $f(t_i)$. - If $\delta_i = 0$, the event time is not observed; we only know $T_i > t_i$, so the contribution is $S(t_i)$.

Thus, the likelihood contribution for observation i can be written as:

$$L_i = f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}.$$

Substituting the exponential forms:

$$L_i = (\lambda_i e^{-\lambda_i t_i})^{\delta_i} \cdot (e^{-\lambda_i t_i})^{1-\delta_i}.$$

Simplifying:

$$L_i = \lambda_i^{\delta_i} e^{-\lambda_i t_i \delta_i} \cdot e^{-\lambda_i t_i (1-\delta_i)} = \lambda_i^{\delta_i} e^{-\lambda_i t_i}.$$

Hence, the full likelihood is:

$$L = \prod_{i=1}^n \lambda_i^{\delta_i} e^{-\lambda_i t_i}.$$

Taking logs:

$$\ell = \sum_{i=1}^n (\delta_i \log \lambda_i - \lambda_i t_i).$$

Now assume a log-linear model for the hazard:

$$\lambda_i = \exp(x_i^\top \beta).$$

$$\ell = \sum_{i=1}^n (\delta_i x_i^\top \beta - t_i \exp(x_i^\top \beta)).$$

Define:

$$\mu_i = t_i \lambda_i = t_i \exp(x_i^\top \beta),$$

which can be interpreted as the expected number of events over exposure time t_i under a constant hazard λ_i .

Then:

$$\begin{aligned} \ell &= \sum_{i=1}^n (\delta_i \log \lambda_i - \mu_i) = \sum_{i=1}^n [\delta_i (\log \mu_i - \log t_i) - \mu_i]. \\ \ell &= \sum_{i=1}^n (\delta_i \log \mu_i - \mu_i) - \sum_{i=1}^n \delta_i \log t_i. \end{aligned}$$

The second term does not depend on β , hence the likelihood is equivalent (up to an additive constant) to:

$$\ell \propto \sum_{i=1}^n (\delta_i \log \mu_i - \mu_i),$$

which is the log-likelihood of a Poisson model with mean $\mu_i = t_i \exp(x_i^\top \beta)$ and observed counts δ_i , with $\log t_i$ acting as an offset.

Coefficients are first estimated on the AFT scale and then converted to hazard ratios using

$$\text{HR} = e^{-\beta}.$$

This is why we report both model coefficients (for inference) and hazard ratios (for practical risk comparison).

Table 4: Coefficient Table with Hazard Ratios and 95% CI

Variable	Value	Std. Error	z	p	H Ratio	lower(95%)	upper(95%)
(Intercept)	-1.6538	2.2450	-0.737	0.4613	5.227	0.064	425.858
age	0.0636	0.0394	1.617	0.1058	0.938	0.869	1.014
treatmentNewDrug	1.2172	0.5921	2.056	0.0398	0.296	0.093	0.945

Interpretation (compact):

- Age: HR = 0.938 per 1-year increase; direction suggests lower hazard with age in this sample, but $p = 0.11$ is not strong at 5%.
- Treatment: HR = 0.296 for NewDrug vs Standard; hazard is lower for NewDrug, with statistical support ($p = 0.04$).
- Global model test ($p = 0.026$): predictors jointly improve fit over intercept-only.

Predictions are generated to convert coefficients into patient-level risk quantities (hazard and probabilities at clinically useful horizons).

Table 5: Predicted risk summary

Age	Treatment	Hazard/month	Survival (6m)	Event Prob. (12m)	Median Survival (months)
50	Standard	0.2168	0.2723	0.9259	3.20
60	NewDrug	0.0340	0.8156	0.3348	20.41
65	NewDrug	0.0247	0.8622	0.2566	28.05

So, coefficients tell us *direction/significance*, hazard ratios tell us *relative risk*, and predictions tell us *absolute risk and expected survival for specific patients*.

5 Probability & Likelihood

5.1 Probability

Definition: Probability quantifies the likelihood of a particular outcome given a known model or set of parameters. It answers:

“What is the chance of observing this outcome, assuming the model and its parameters are true?”

Perspective: Forward-looking. Start with model parameters and compute the probability of data.

Mathematical Formulation:

$$P(X | \theta)$$

where:

- X : Observed data or event
- θ : Parameters of the model

Example: For a fair coin with $P(\text{Head}) = 0.5$, the probability of observing 3 heads in 5 flips is computed using the binomial distribution.

5.2 Likelihood

Definition: Likelihood measures how plausible a set of model parameters is given the observed data. It answers:

“How well does this set of parameters explain the observed data?”

Perspective: Backward-looking. Start with observed data and evaluate different parameter values.

Mathematical Formulation:

$$L(\theta | X) = P(X | \theta)$$

Conceptually, X is fixed and θ varies.

Example: Given 3 heads in 5 flips, likelihood helps determine the value of $\theta = P(\text{Head})$ that best explains the observed data.

5.3 Key Differences

Aspect	Probability	Likelihood
Purpose	Predict future data/events	Estimate model parameters
Input	Model parameters (θ)	Observed data (X)
Output	Chance of observing data	Plausibility of parameters
Role in Inference	Forward problems	Parameter estimation (MLE)
Fixed vs Variable	θ fixed, X varies	X fixed, θ varies

5.4 Maximum Likelihood Estimation (MLE)

Concept: MLE aims to find parameter values that maximize the likelihood of the observed data.

Likelihood Function:

$$L(\theta | X) = \prod_{i=1}^n f(X_i | \theta)$$

MLE Estimator:

$$\hat{\theta} = \arg \max_{\theta} L(\theta | X)$$

5.5 Log-Likelihood

To simplify computation, take logarithms:

$$\ell(\theta | X) = \log L(\theta | X) = \sum_{i=1}^n \log f(X_i | \theta)$$

Maximizing $\ell(\theta | X)$ is equivalent to maximizing $L(\theta | X)$.

5.6 Steps in MLE

1. Specify the probabilistic model
2. Write the likelihood function
3. Compute the log-likelihood
4. Differentiate and solve:

$$\frac{\partial \ell(\theta | X)}{\partial \theta} = 0$$

5.7 Example: Normal Distribution

Assume $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$.

Log-Likelihood:

$$\ell(\mu, \sigma^2 | X) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

MLE Estimates:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

5.8 Properties of MLE

- Consistency: $\hat{\theta} \rightarrow \theta$ as $n \rightarrow \infty$
- Asymptotic Normality
- Efficiency under regularity conditions

5.9 Challenges

- Non-convex optimization landscapes
- Lack of closed-form solutions
- Overfitting in high-dimensional settings

5.10 Applications

MLE underpins:

- Logistic Regression
- Gaussian Mixture Models
- Hidden Markov Models
- Neural Networks (via negative log-likelihood)

5.11 Further Directions

- Maximum A Posteriori (MAP) Estimation
- Numerical Optimization (Newton-Raphson, Gradient Ascent, EM)
- Bayesian Inference